UNIT - I : INTRODUCTION TO INDUSTRIAL ENGINEERING,

I. Industrial Engineering (I.E.)

Definition

* Industrial Engineering is concerned with the DESIGN, IMPROVEMENT and INSTALLATION of INTEGRATED SYSTEMS of PEOPLE, MATERIALS, INFORMATION, EQUIPMENT and ENERGY. It draws on knowledge of mathematical, physical and social sciences together with the principles of engineering analysis and design to specify, predict and evaluate the results from such systems. (Institute of Industrial Engineers, USA - IISE)

Simple meaning: I.E. = doing the same work with fewer resources - increasing PRODUCTIVITY by eliminating waste of man, material, machine, money and time.

Development of I.E.

Period	Contribution
1776	Adam Smith - 'Wealth of Nations': division of labour
1798	Eli Whitney - interchangeable parts
1832	Charles Babbage - study of time and cost of operations
1880-1911	F. W. Taylor - Scientific Management, time study (father of I.E.)
1900-1912	Frank and Lillian Gilbreth - motion study, therbligs
1910s	Henry Gantt - Gantt chart, task and bonus system; Henry Ford - assembly line
1920s-30s	Elton Mayo - Hawthorne studies; Shewhart - quality control charts
1940s+	Operations research, linear programming, computers, Deming and Juran (TQM)
Today	Lean, Six Sigma, ERP, supply chain, simulation, Industry 4.0

Applications of I.E.

* Manufacturing industries (plant layout, line balancing, production planning and control), service sector (hospitals, banks, airlines, retail), construction, defence, logistics and transport, software and IT, government.
* Typical activities: method study, work measurement, facility location and layout, inventory control, quality control, maintenance planning, cost reduction and value analysis, wage and incentive systems, materials handling, project scheduling (CPM / PERT), safety, ergonomics.

Role of an industrial engineer

* Analyses existing systems and finds the BEST method; designs new facilities / layouts; sets standards of time and cost; improves

productivity and quality; plans manpower, materials and machines; reduces cost and waste; acts as a link between top management and the shop floor (staff function advising line managers); introduces new technology and trains people.

Quantitative tools of I.E.

* Operations Research (linear programming, transportation, assignment, queuing, simulation, network analysis - CPM / PERT), statistics and probability (SQC, forecasting, regression), work study and work measurement, engineering economy, inventory models (EOQ), computer aided design and ERP, decision theory, line balancing, learning curves.

Productivity measurement

* PRODUCTIVITY = relationship between OUTPUT obtained and INPUT used. Increase by producing more with same input or same output with less input.

Productivity = Output / Input

* Partial productivity: labour productivity = units produced / man-hours; machine productivity = output / machine hour; material productivity = output / kg of material. Total productivity = total output value / total input value (labour + material + capital + energy + other).

Type	Formula	Example
Labour	Output / labour hours	500 units / 100 h = 5 units / h
Material	Output / material consumed	500 units / 250 kg = 2 units / kg
Total factor	Output value / total input value	Rs 50,000 / Rs 40,000 = 1.25

* Productivity index = (productivity in current period / productivity in

base period) x 100. Factors affecting: technology, work methods, layout, quality of labour, management, motivation, working conditions, raw material quality.

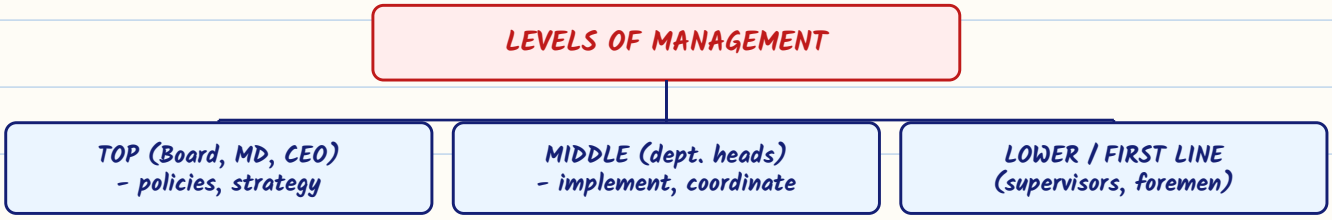
2. Concepts of Management

* MANAGEMENT = the art and science of getting things done through people in formally organised groups, by planning, organising, staffing, directing and controlling resources to achieve organisational goals efficiently and effectively. (Koontz: 'the art of getting things done through and with people in formally organised groups').
* Management is: a PROCESS (series of functions), a SCIENCE (organised knowledge), an ART (skill in application), a PROFESSION, multi-disciplinary, universal, goal oriented, a group activity and continuous.

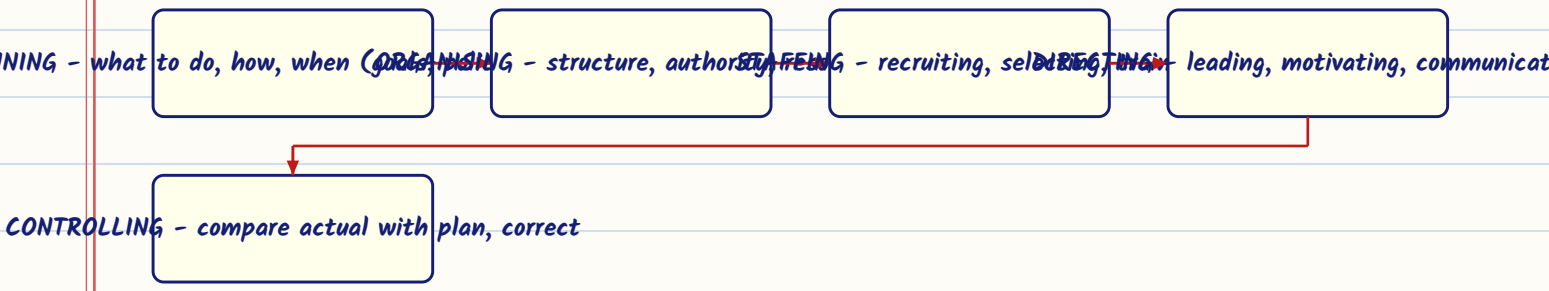
Importance of management

* Achieves group goals; increases efficiency (cuts cost); creates a dynamic organisation which adapts to change; achieves personal objectives; ensures optimum use of resources (5 M's: Men, Machines, Materials, Money, Methods); social benefits - employment, quality goods, welfare; gives stability, growth and competitive advantage.

Levels of management



Functions of management



* PLANNING is the primary function (looking ahead); ORGANISING arranges tasks and relations; STAFFING (HR) puts the right man on the right job; DIRECTING (actuating) guides and motivates; CONTROLLING measures performance against standards and takes corrective action; COORDINATION is the essence of all functions. (POSDCORB - Gulick: Planning, Organising, Staffing, Directing, Coordinating, Reporting, Budgeting).

3. Scientific Management

* Developed by FREDERICK WINSLOW TAYLOR (1856 - 1915, 'Father of Scientific Management'), an American engineer at Midvale and

Bethlehem Steel. Book: 'The Principles of Scientific Management' (1911).

* Definition: 'Knowing exactly what you want men to do and seeing that they do it in the best and cheapest way.' It replaces rule-of-thumb methods by SCIENTIFIC investigation of each task, aiming at maximum output with minimum effort, and prosperity for both employer and employee.

Techniques of Taylor

* Time study, motion study, fatigue study, standardisation of tools and methods, differential piece rate wage system, functional foremanship (8 specialised foremen: 4 in planning - route clerk, instruction card clerk, time and cost clerk, shop disciplinarian; 4 in execution - gang boss, speed boss, repair boss, inspector), scientific selection and training, mental revolution.

Taylor's principles of scientific management

No.	Principle	Meaning
1	Science, not rule of thumb	Study every job scientifically to find the one best way
2	Harmony, not discord	Cooperation between management and workers; no conflict
3	Cooperation, not individualism	Teamwork, mutual help, replacing chaos
4	Maximum output, not restricted output	Maximum production in place of restricted output - prosperity for all
5	Development of each man to his greatest efficiency	Scientific selection, training and development of workers

* Mental revolution: management and workers must change attitude -

both should focus on increasing the surplus (not on dividing it).

* Criticisms: treats workers like machines; ignores human and social needs; monotony, over-specialisation; opposed by trade unions; functional foremanship violates unity of command; applies mostly at shop-floor level.

4. Fayol's Principles of Management

* HENRI FAYOL (1841 - 1925), French mining engineer, 'Father of Modern (Administrative / General) Management'. Book: 'Administration Industrielle et Generale' (1916). Divided industrial activities into: technical, commercial, financial, security, accounting, managerial.

14 principles of management

No.	Principle	Idea
1	Division of work	Specialisation increases efficiency
2	Authority and responsibility	Right to give orders must be matched by responsibility
3	Discipline	Obedience, respect for rules and agreements
4	Unity of command	One employee, ONE boss
5	Unity of direction	One head, one plan for a group of activities with same objective
6	Subordination of individual interest to general interest	Organisation's goal first
7	Remuneration	Fair wages satisfying both employer and employee
8	Centralisation	Degree to which decision making is concentrated at the top
9	Scalar chain	Line of authority from top to bottom (Fayol's gang plank = direct link)
10	Order	A place for everything and everything in its place (material and social order)
11	Equity	Kindliness and justice to employees
12	Stability of tenure of personnel	Avoid unnecessary turnover
13	Initiative	Freedom to think out and execute a plan
14	Esprit de corps	Union is strength - team spirit; harmony

Taylor	Fayol
Scientific management; shop floor level (operational)	General administrative theory; top management level
Focus on efficiency of workers and methods	Focus on managerial functions and principles
Functional foremanship (violates unity of command)	Unity of command (one boss)
Principles developed by observation and experiment	Principles developed from personal experience
Called 'Father of Scientific Management'	Called 'Father of Modern Management'

5. Plant Location and Layout

Factors governing plant location

* Plant location = selecting the geographical site for a factory - a long term decision affecting cost, competitiveness and growth. Two steps: choosing the region, then the exact site.

Factor	Considerations
Nearness to raw materials	Transport cost, availability, quality (steel, cement, sugar plants)
Nearness to market	Perishable / bulky products; distribution cost, customers
Transport facilities	Road, rail, port and air connectivity
Power, water, fuel	Cost and reliability of electricity and water; effluent disposal
Labour supply	Skilled / unskilled availability, wage rates, labour relations
Climate and environment	Humid for textiles, dry for precision instruments
Land, site cost, soil	Price, drainage, future expansion, soil bearing capacity
Government policy	Subsidies, tax holidays, industrial estates, SEZ, pollution control rules
Other	Banking and insurance, housing, schools, hospitals, community attitude, safety, competitors

* Quantitative methods: weighted factor rating, break-even (cost-volume) analysis, transportation model, centre of gravity method (x = sum(w x)/sum(w)).

Centre of gravity: $X = \text{sum}(w_i x_i) / \text{sum}(w_i)$, $Y = \text{sum}(w_i y_i) / \text{sum}(w_i)$

Plant layout

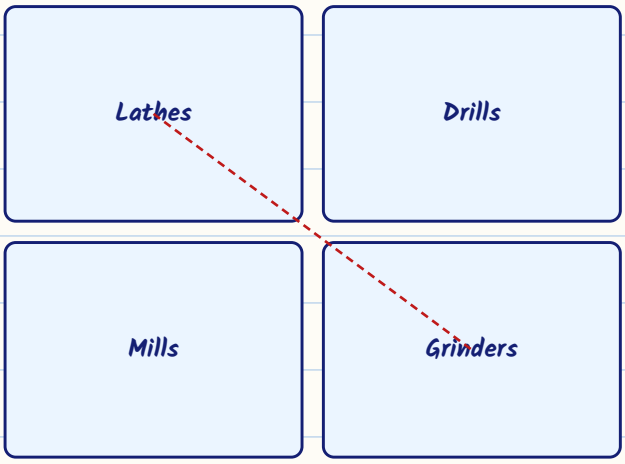
* PLANT LAYOUT = the physical arrangement of machines, equipment, materials, work centres and service areas within a building for the most efficient flow of material and people. Objectives: minimum material handling, smooth flow, minimum production cycle time, efficient use of floor space, safety and comfort, flexibility for future

change.

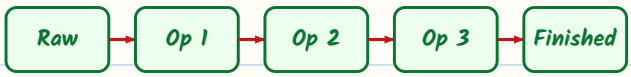
Types of production layouts

* 1. PROCESS (functional) layout - similar machines grouped by function (all lathes together, all drills together). 2. PRODUCT (line) layout - machines arranged in the sequence of operations. 3. FIXED POSITION (static) layout - product stays at a location, men and machines move to it (ships, aircraft, bridges, dams, turbines). 4. COMBINATION (hybrid) layout. 5. GROUP TECHNOLOGY / CELLULAR layout - machines grouped into cells to make a family of similar parts.

PROCESS LAYOUT



PRODUCT (LINE) LAYOUT



Process layout	Product layout
Machines grouped by type / function	Machines in order of operations
General purpose machines, flexible	Special purpose machines, rigid
Low volume, many varieties (job / batch)	High volume, few varieties (mass / continuous)
Skilled labour needed	Semi-skilled labour is enough
High work-in-process, long cycle time	Low work-in-process, short cycle time
More material handling, backtracking	Minimum material handling, straight flow
Breakdown of one machine does not stop all	Breakdown stops the whole line
Low investment per machine, high handling cost	High investment, low unit cost

Advantages and disadvantages

	Process layout	Product layout
Advantages	Flexible, handles variety; lower investment; machine breakdown not critical; better utilisation; job satisfaction, easy supervision of skills	Smooth flow, low handling; low WIP and cycle time; simple control and inspection; less floor space; low unit cost; unskilled labour
Disadvantages	High handling and WIP cost; complex scheduling; long production time; more floor space and inventory; skilled workers needed	Inflexible to product change; heavy investment; one breakdown stops line; monotony; line balancing needed; low volume uneconomical
Applications	Machine shops, hospitals, tool rooms, repair shops, universities	Automobile assembly, bottling, cement, sugar, refineries, TV and appliances

Quantitative techniques for optimal design of layouts

* 1. TRAVEL CHART (from-to chart): matrix of trips / loads between departments; departments with high flow are placed close together. Objective: minimise total material handling cost = sum of (flow x distance x cost per unit distance).
* 2. Relationship (REL) chart (Muther): closeness ratings A (absolutely necessary), E, I, O, U (unimportant), X (undesirable). 3. CRAFT (Computerised Relative Allocation of Facilities Technique) improves an existing layout by exchanging departments; ALDEP, CORELAP construct layouts. 4. Line balancing (assembly line): assign tasks to stations to minimise idle time.

Cycle time C = available time / required output; Min. stations = (sum of

Eg: tasks total 24 min, cycle time 6 min -> minimum stations = 24/6 = 4. If 5 stations are used, efficiency = 24/(5 x 6) = 80%.

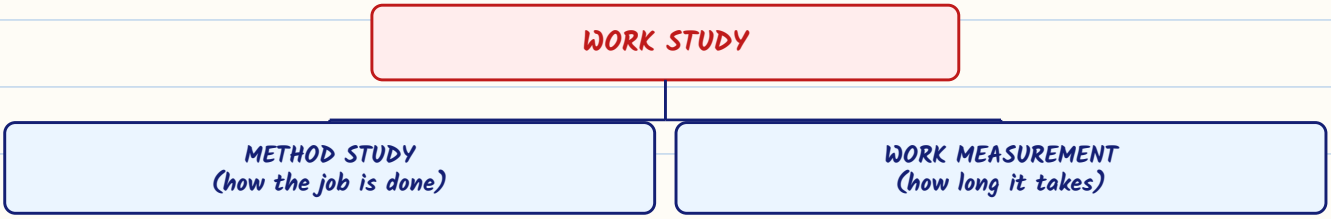
* 5. Queueing and simulation models, linear programming and quadratic assignment formulation of layout; Systematic Layout Planning (SLP): flow + activity relationship -> space -> layout.

Revision: IE = design of integrated systems; Taylor - 5 principles (science, harmony, cooperation, max output, development); Fayol - 14 principles; layouts: process, product, fixed position, cellular.

UNIT - II : WORK STUDY

I. Introduction and importance

* WORK STUDY = systematic examination of the methods of carrying out activities so as to improve effective use of resources and to set up standards of performance for the activities carried out (ILO). It is the foundation of productivity improvement and is also called 'Methods Engineering'.
* Two parts: METHOD STUDY (motion study) - finds the best way of doing a job; WORK MEASUREMENT (time study) - finds the time a qualified worker takes to do it by the best method. Method study first, then work measurement.



Importance of work study

* Increases productivity without capital investment; lowers cost; gives standard times for production planning, scheduling and costing; basis for wage incentive plans; eliminates waste and unnecessary movement; better layout and material handling; reduces fatigue, improves safety and working conditions; fair workload and job satisfaction.

Types of production and application of work study

Type	Features	Work study use
Job production	One / few items to customer order, general purpose machines, skilled labour	Less detailed; method study on important jobs
Batch production	Moderate quantity in lots, repeated at intervals	Time standards for each batch operation
Mass (flow) production	Large volume of standard product, special purpose machines	Detailed method and time study; line balancing
Continuous (process) production	Non stop, chemicals, oil, cement, sugar	Process analysis, work sampling

* Work study applies to manufacturing, offices (clerical work), hospitals, banks, maintenance, construction, warehouses, hotels and retail.
* PROCEDURE (SREDIM): Select the job, Record all facts, Examine critically, Develop the best method, Install it, Maintain it.

2. Method Study

* METHOD STUDY = systematic recording and critical examination of existing and proposed ways of doing work, as a means of developing and applying easier and more effective methods and reducing costs.

Steps (SREDIM)

Step	Action
S - Select	Choose the job: high cost, bottleneck, high labour content, repetitive, hazardous, complaints
R - Record	Record all relevant facts by charts, diagrams, films
E - Examine	Critical examination: ask Purpose? Place? Sequence? Person? Means? -> Why? ; Eliminate, Combine, Rearrange, Simplify (ECRS)
D - Develop	Develop practical, economical, effective method
I - Install	Get approval, train workers, put into practice
M - Maintain	Follow up, check that the method is not reverted

Recording techniques – Process charts and symbols



Operation Inspection Transport Delay Storage

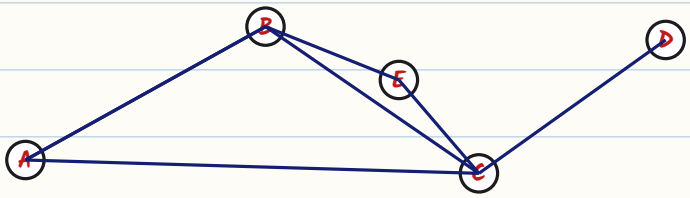
* OUTLINE PROCESS CHART: only operations and inspections (overall picture). FLOW PROCESS CHART: all five symbols - operation, inspection, transport, delay, storage; three types - MAN type (what the worker does), MATERIAL type (what happens to material), EQUIPMENT type. Includes distance moved and time.

Symbol	Meaning	Eg.
Circle	Operation - changes the object (productive)	Turning a shaft
Square	Inspection - check quality or quantity	Gauging diameter
Arrow	Transport - move from one place to other	Carry by trolley
D (big)	Temporary delay - waiting	Waiting in queue
Inverted triangle	Storage - controlled keeping	Raw material store
Circle in square	Combined operation + inspection	Machine and gauge

Only OPERATION is value-adding; transport, delay, storage, inspection are non-value-adding - to be eliminated or reduced by method study.

Flow diagram and string diagram

* FLOW DIAGRAM: scale plan of the shop on which the path of movement of the worker / material is drawn with the process chart symbols. STRING DIAGRAM: a scale model of the layout; a thread is passed round pins at work points to show the movement paths; the length of thread measures the total travel. Shows congestion and backtracking.



String diagram: scale layout of the shop; a thread is wound round pins at work points to trace movement paths of worker / material. Most thread between two points -> place them nearer.

Example: flow process chart (man type) - drilling a component

Step	Description	Symbol	Dist (m)	Time (min)
1	Take casting from bin	Operation	-	0.20
2	Carry to drilling machine	Transport	12	0.40
3	Wait for machine	Delay	-	0.50
4	Clamp and drill holes	Operation	-	1.80
5	Check hole diameter	Inspection	-	0.30
6	Carry to store	Transport	20	0.60
7	Store in tray	Storage	-	-

Summary: operations 2, transport 2 (32 m), inspection 1, delay 1, storage 1. Improvement by ECRS: place machine nearer bin (cut transport), keep a second casting ready (remove delay).

* Critical examination - PRIMARY questions: What is done? Why? Where? Why there? When? Why then? Who does it? Why that person? How? Why that way? -> SECONDARY questions then ask: what else could be done, where else, when else, who else, how else -> choose what SHOULD be done.

Other recording tools

* MULTIPLE ACTIVITY (man-machine) chart: time scale chart showing activities of several workers / machines to find idle time (man-machine ratio, number of machines per operator). SIMO chart (simultaneous motion cycle): left and right hand motions of a worker on a common time scale using therbligs. Travel chart (from-to chart), cyclegraph (photograph of light on moving hand), chronocyclegraph (interrupted light - speed), film analysis (memo-motion, micro-motion).

Motion study and the principles of motion economy

* MOTION STUDY (Gilbreths) = analysis of the body motions used in doing a job to eliminate wasted motion and set up the best motion sequence. Frank Gilbreth identified 17 basic divisions of human motion called THERBLIGS (Gilbreth spelt backwards, with th transposed).

Therbligs

Therblig	Symbol	Class	Therblig	Symbol	Class
Search	Sh	Ineffective	Hold	H	Ineffective
Select	St	Ineffective	Rest for overcoming fatigue	R	Ineffective
Grasp	G	Effective	Unavoidable delay	UD	Ineffective
Transport empty	TE	Effective	Avoidable delay	AD	Ineffective
Transport loaded	TL	Effective	Plan	Pn	Ineffective
Position	P	Effective	Inspect	I	Effective
Assemble	A	Effective	Release load	RL	Effective
Disassemble	DA	Effective	Pre-position	PP	Effective
Use	U	Effective			

* Aim: eliminate ineffective therbligs (search, select, plan, hold, delays, rest) and shorten the effective ones.

Principles of motion economy (Barnes)

* (a) Use of the HUMAN BODY: both hands begin and end motions together; hands not idle together except rest; motions of arms symmetrical and in opposite directions; use the lowest classification of body movement (finger < wrist < forearm < upper arm < shoulder / body); momentum helps; smooth continuous curved motion preferred over zig-zag; rhythm.
* (b) ARRANGEMENT of the WORKPLACE: fixed place for tools and

materials; tools pre-positioned; gravity feed bins and drop delivery; materials and tools in order of sequence; proper lighting, seating, height of work.

* (c) Design of TOOLS and EQUIPMENT: use jigs and fixtures instead of hands for holding; combine two or more tools; foot operated devices; handles designed for maximum contact; levers and wheels within easy reach.

Micro-motion study

* Detailed study of short-cycle repetitive operations by filming the work at constant speed (e.g. 1000 frames per minute / 16-24 fps) along with a MICROCHRONOMETER (clock) in the picture; the film is analysed frame by frame to break the job into therbligs and time each element in WINKS (1 wink = 1/2000 min). Used for highly repetitive, mass production work; costly. Memo-motion: filming at slow speed (1 frame / second) for long cycle jobs.

3. Work Measurement (Time Study)

* WORK MEASUREMENT = application of techniques to establish the time for a QUALIFIED worker to carry out a specified job at a defined level of performance. Uses: production planning, labour cost, incentive plans, manpower and machine requirement, delivery dates, comparing methods.
* Techniques: (1) time study (stopwatch), (2) work sampling (ratio delay), (3) synthesis from standard data, (4) predetermined motion time systems (PMTS), (5) analytical estimating, (6) historical records.

Time study procedure

* 1. Select the job and worker (average qualified worker). 2. Record information on the study sheet. 3. Break the job into ELEMENTS (short, identifiable). 4. Time each element with a stopwatch (cumulative / snap-back / continuous method) for a number of cycles. 5. RATE the worker's performance. 6. Compute the normal time. 7. Add ALLOWANCES to get the standard time. 8. Check and publish.
* Number of observations required: n = (40 sqrt(N sum(x^2) - (sum x)^2) / sum(x))^2 at 95% confidence and +- 5% accuracy (N = number of readings taken).

Observed time (OT) = average of selected cycle times

Normal Time (NT) = OT x Rating factor / 100

Standard Time (ST) = NT + Allowances = NT x (1 + allowance fraction)

If allowances are x% of the TOTAL (standard) time: ST = NT / (1 - x)

* ALLOWANCES: personal needs (5-7%), basic fatigue (4%), variable fatigue (conditions of work), special / process allowance, contingency, delay allowance, policy allowance. Total typically 15 - 20%.

Rating techniques

* RATING = assessing the speed (tempo) of the worker relative to the NORMAL (standard) pace which is defined as 100 (60 on the British 0-100 scale; walking 4 miles per hour on level ground, dealing 52 cards in 0.5 minute).

Method	Idea
Speed rating	Rate the pace alone against normal 100 (e.g. 120 for a fast worker)
Westinghouse (levelling) system	Four factors: skill, effort, conditions, consistency, each graded (superskill +0.15 ... poor -0.22); R = 1 + sum of factors
Synthetic rating (Morrow)	Compare element times with PMTS (e.g. MTM): R = PMTS time / actual time
Objective rating (Mundel)	Pace rating + job difficulty adjustment factor
Physiological	Measures heart rate / oxygen consumption

Eg: Observed time of an element = 0.80 min, rating = 120, allowance = 15% (of normal time). NT = 0.80 x 1.20 = 0.96 min; ST = 0.96 x 1.15 = 1.104 min; standard output per 8-hour day = 480/1.104 = 435 pieces.
Eg 2: if the 15% allowance is on the total (standard) time: ST = 0.96 / (1 - 0.15) = 1.129 min.

Numerical example - time study

A job has 4 elements. Average observed times (min) and ratings are given. Allowances = 12% of normal time.

Element	Avg. observed (min)	Rating	Normal time
1	0.50	100	0.500
2	0.80	110	0.880
3	0.30	90	0.270
4	0.60	120	0.720
Total			2.370

Standard time = 2.370 x 1.12 = 2.654 min per piece

Output per 8 hour shift = 480 / 2.654 = 180.9, i.e. about 181 pieces.

Work sampling (ratio delay study)

* Statistical technique of making a LARGE number of RANDOM instantaneous observations of a worker / machine over a period to find the proportion of time spent in each activity (working, idle, delay). Developed by L.H.C. Tippett (UK, textile mills, 1934) - 'snap reading method'.
* Advantages: no stopwatch, many workers can be studied by one observer, less costly, no continuous watching, can be interrupted, good for non-repetitive jobs and idle time of machines. Disadvantages: gives no details of method, less suitable for short cycle jobs, observer error.

Sample size $N = 4\,p\,(1-p) / (L)^2$ (95% confidence, L = absolute accura

Eg: p = 0.30 (30% idle), required accuracy +-5% -> N = 4 (0.3)(0.7) / (0.05)^2 = 336 observations.

Standard time = (total time x % working x rating) / number of units produced x (1 + allowance).

Predetermined Motion Time Systems (PMTS)

* PMTS = techniques where STANDARD times (established from film studies) for basic human MOTIONS (reach, move, grasp, position, release, body motions) are used to build up the time for a job without stopwatch - the time is predetermined from tables of the motions involved. Times in TMU (time measurement unit): 1 TMU = 0.00001 hour = 0.0006 min = 0.036 second. Used even at the planning (design) stage, and no rating is needed.
* Systems: Work Factor (WF), Methods Time Measurement (MTM-1,

MTM-2, MTM-3), Basic Motion Time Study (BMT), Dimensional Motion Times (DMT), Master Standard Data, Modular Arrangement of PMTS (MODAPTS), Maynard operation sequence technique (MOST).

Methods Time Measurement (MTM)

* Developed by H. B. Maynard, G. J. Stegemerten and J. L. Schwab (Westinghouse, 1948). Each motion - REACH (R), MOVE (M), TURN (T) and APPLY PRESSURE, GRASP (G), POSITION (P), RELEASE (RL), DISENGAGE (D), eye travel and eye focus, body, leg and foot motions, crank (C), is analysed by DISTANCE moved, CASE (type), and WEIGHT, and its time is read from a table in TMU. 1 TMU = 0.00001 hour.
* Eg: Reach 20 cm case B -> about 12 TMU; Move 20 cm with 2 kg -> 13 TMU. Total TMU x 0.0006 = minutes; standard time includes allowance.

Work Factor system

* Developed by Quick, Shea and Koehler (RCA, 1938). Time for a motion depends on body member used (finger, hand, forearm, arm, trunk, leg, foot), DISTANCE moved, and WORK FACTORS - extra time for difficulty: (1) manual control - steering (S), directional control (U), precaution (P), definite stop (D), change of direction (C); (2) weight / resistance. Time in work-factor units; 1 WF unit = 0.0001 min (detailed) in Detailed WF; Ready WF, Abbreviated WF, Brief WF versions.

MTM	Work Factor
Motion analysed by distance, case, weight	By body member, distance, number of work factors
Unit TMU = 0.0006 min	Unit = 0.0001 min (detailed)
Rating not needed (tables = normal)	Rating inherent in the tables
More widely used	Developed by RCA; used for light assembly

4. Ergonomics

* ERGONOMICS (Human factors engineering; Greek ergon = work, nomos = law) = science of fitting the JOB to the WORKER - the study of the interaction between man, machine and working environment to improve comfort, efficiency, health and safety.

Principles of ergonomics

* 1. Match the workstation, tools and controls to the body dimensions (ANTHROPOMETRY - 5th to 95th percentile of the population). 2. Keep the posture neutral; avoid awkward, static, twisted or repetitive postures. 3. Reduce excessive force; use mechanical aids. 4. Keep everything in easy REACH - normal and maximum work areas. 5. Provide adequate clearance, adjustable seats and benches.
* 6. Control the ENVIRONMENT: lighting (glare free, 300 - 500 lux for assembly), noise (< 85 dB), temperature and humidity, ventilation, vibration. 7. Display design: clear, readable dials, colour coding, compatibility between control movement and display response. 8. Minimise fatigue: work-rest schedule, job rotation, task variety. 9. Safety: guards, emergency stops, fail-safe design. 10. Involve users:

participatory ergonomics.

* Areas: workstation design (sitting / standing), man-machine system, display and controls, lifting and manual handling (NIOSH lifting equation), cognitive ergonomics, office ergonomics (VDU, chair, keyboard). Benefits: higher productivity and quality, fewer accidents, less absenteeism and musculoskeletal disorders (back pain, carpal tunnel syndrome).

Anthropometry and workplace design

Item	Recommended
Normal work area (arm sweep, elbow fixed)	About 25 - 40 cm radius (hand); maximum area 60 - 70 cm
Sitting work surface height	About 70 - 75 cm (elbow height) for light work
Standing work height	Heavy work 10 - 15 cm below elbow; precision work 5 - 10 cm above
Seat	Adjustable height, lumbar (back) support, foot rest
Illumination (lux)	General 150, assembly 300 - 500, fine inspection 1000+
Noise	Below 85 dB(A) for 8 h exposure

Comparison of method study and work measurement

Method study	Work measurement
Finds the BEST METHOD of doing the job	Finds the TIME for the best method
Done first	Done after the method is standardised
Tools: charts, diagrams, SREDIM	Tools: stopwatch, work sampling, PMTS
Aim: eliminate waste motion	Aim: set time standard

Important questions

* 1. Define work study; explain method study procedure with SREDIM.
* 2. Explain flow process chart symbols; differentiate flow diagram and string diagram.
* 3. What are therbligs? Explain principles of motion economy.
* 4. Explain time study procedure; rating techniques; compute standard time.
* 5. Explain work sampling and its merits; PMTS, MTM and work factor system.
* 6. Explain principles of ergonomics.

Revision: Work study = method study + work measurement. SREDIM. ST = NT + allowance, NT = OT x rating. Work sampling - random observations. PMTS (MTM, WF) - TMU. Therblig - 17 basic motions.

UNIT - III : STATISTICAL QUALITY CONTROL AND TOTAL

I. Quality and Quality Control

* QUALITY = fitness for use (Juran); conformance to requirements (Crosby); the totality of features and characteristics of a product or service that bear on its ability to satisfy stated or implied needs (ISO). Dimensions (Garvin): performance, features, reliability, conformance, durability, serviceability, aesthetics, perceived quality.
* QUALITY CONTROL (QC) = the operational techniques and activities used to fulfil requirements for quality - inspection, testing and corrective action to ensure that products meet specifications at least cost.
* QUALITY ASSURANCE (QA) = planned and systematic activities (the system - procedures, audits, documentation) implemented to GIVE CONFIDENCE that the quality requirements will be met. QC is product oriented (detects defects); QA is process / system oriented (prevents defects).

Quality Control	Quality Assurance
Detection - correct defects	Prevention - avoid defects
Product oriented	Process / system oriented
Line function (inspection, testing)	Staff function (audits, standards)
Reactive	Proactive

Importance of quality assurance

* Customer satisfaction and loyalty; reputation and competitiveness; lower cost (less scrap, rework, warranty claims); higher productivity; legal and safety compliance; access to markets (ISO certification);

reduced liability; employee morale.

* Cost of quality: PREVENTION cost (training, planning), APPRAISAL cost (inspection, testing), INTERNAL failure (scrap, rework), EXTERNAL failure (returns, warranty, lost goodwill). Spend more on prevention - total cost falls.

Inspection

* Checking the product against specification. 100% (complete) inspection - for critical parts, small lots, costly and boring, fatigue errors, destructive tests impossible. SAMPLING inspection - for large lots, destructive testing, cheaper, faster. Types: incoming, in-process, final; first piece, floor, centralised.

2. Statistical Quality Control (SQC)

* SQC = application of STATISTICAL methods (probability theory, sampling) to control and improve quality; started by W. A. SHEWHART (Bell Labs, 1924, control chart); sampling plans by Dodge and Romig. Two branches: (1) Process control - CONTROL CHARTS, (2) Product control - ACCEPTANCE SAMPLING.

Causes of variation

* No two items are exactly alike. CHANCE (random / common) causes: many small unavoidable causes, always present, predictable pattern - process is 'in statistical control'. ASSIGNABLE (special) causes: definite, identifiable (worn tool, bad material, operator error), cause big variation - must be found and removed.

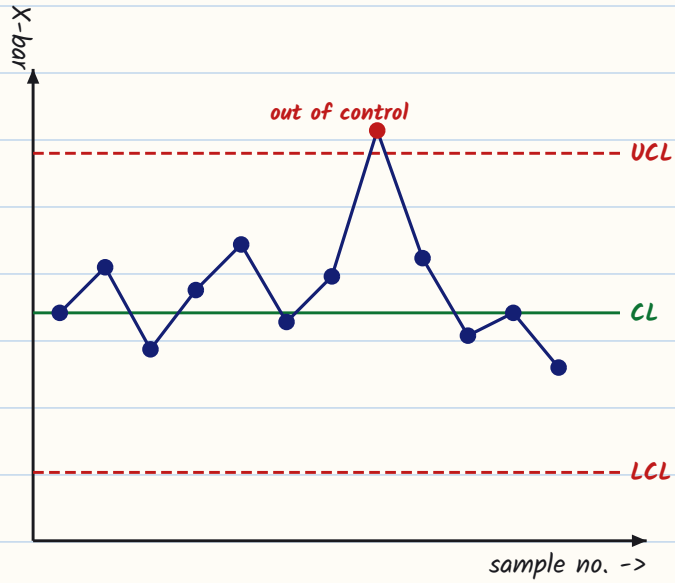
Statistical basis

* Quality characteristic follows NORMAL distribution: mean mu, standard deviation sigma. Within mu +- 3 sigma lies 99.73% of items (mu +- 1

sigma 68.27%, +- 2 sigma 95.45%). Hence control limits are set at +- 3 sigma ('three sigma limits'): the chance of a point outside by chance alone is only 0.27%.

* Variables = measurable characteristics (length, weight, temperature) -> X-bar, R, s charts. Attributes = go / no-go characteristics (defective or not, number of defects) -> p, np, c, u charts.

3. Control Charts for Variables



* Control chart = graph of sample statistic against sample number with a CENTRE LINE (CL), UPPER and LOWER control limits (UCL, LCL). Points inside limits with random pattern -> process in control. Point outside limits, run of 7 on one side, trend, cycles -> out of control: search for assignable cause.
* Procedure: take k samples (usually 20 - 25) each of n items (n = 4 or 5); compute X-bar and R (range = max - min) for every sample; compute grand mean X-double-bar and average range R-bar; compute limits; plot; analyse.

X-bar and R charts

$$\text{X-double-bar} = \text{sum}(\text{X-bar}_i) / k \text{ , R-bar} = \text{sum}(R_i) / k$$

$$\text{X-bar chart: UCL} = \text{X-dbar} + A_2\ \text{R-bar} \text{ , LCL} = \text{X-dbar} - A_2\ \text{R-bar}$$

$$\text{R chart: UCL} = D_4\ \text{R-bar} \text{ , LCL} = D_3\ \text{R-bar}$$

n	A2	D3	D4	d2	A3	B3	B4
2	1.880	0	3.267	1.128	2.659	0	3.267
3	1.023	0	2.574	1.693	1.954	0	2.568
4	0.729	0	2.282	2.059	1.628	0	2.266
5	0.577	0	2.114	2.326	1.427	0	2.089
6	0.483	0	2.004	2.534	1.287	0.030	1.970
7	0.419	0.076	1.924	2.704	1.182	0.118	1.882

* Estimate of process standard deviation sigma = R-bar / d2. The R chart is examined first (variability), then the X-bar chart (central tendency).

X-bar and s charts

$$\text{X-bar chart: UCL} = \text{X-dbar} + A_3\ \text{s-bar} \text{ , LCL} = \text{X-dbar} - A_3\ \text{s-bar}$$

$$\text{s chart: UCL} = B_4\ \text{s-bar} \text{ , LCL} = B_3\ \text{s-bar}$$

* s (sample standard deviation) is used instead of R when sample size is LARGE (n > 10) - range loses efficiency; computation is heavier, but s is a better measure of spread.

Numerical example (X-bar and R)

10 samples of size n = 5 gave sum of sample means = 500.0 and sum of

ranges = 30.0 (A2 = 0.577, D3 = 0, D4 = 2.114).

X-dbar = 500/10 = 50.0 ; R-bar = 30/10 = 3.0

UCL(X) = 50 + 0.577 x 3 = 51.73 ; LCL(X) = 50 - 1.731 = 48.27

UCL(R) = 2.114 x 3 = 6.342 ; LCL(R) = 0 x 3 = 0

Process sigma = 3.0 / 2.326 = 1.29. Specification 50 +- 4: Cp = (USL - LSL)/(6 sigma) = 8/7.74 = 1.03 (just capable).

Applications

* Control of dimensions of machined parts (diameter, thickness), weight / fill of packages, tablet weight in pharmaceuticals, temperature or pH in chemical processes, electrical parameters of components, service times. Used to monitor, detect shifts early, decide when to adjust, establish process capability and reduce scrap.

4. Control Charts for Attributes

Chart	Plots	Sample size	Limits
p chart	Fraction defective p	Variable or constant	p-bar +- 3 sqrt(p-bar (1 - p-bar) / n)
np chart	Number defective	Constant	np-bar +- 3 sqrt(np-bar (1 - p-bar))
c chart	Number of defects per unit	Constant (one unit)	c-bar +- 3 sqrt(c-bar)
u chart	Defects per unit	Variable	u-bar +- 3 sqrt(u-bar / n)

Eg: 20 samples of 100 items had 60 defectives in all: p-bar = 60/2000 =

0.03; UCL = 0.03 + 3 sqrt(0.03 x 0.97 / 100) = 0.03 + 0.0512 = 0.0812; LCL = 0.03 - 0.0512 < 0 -> taken as 0.

c chart eg: average 9 defects per fabric roll: UCL = 9 + 3 x 3 = 18; LCL = 9 - 9 = 0.

5. Acceptance Sampling - Attribute Sampling Inspection

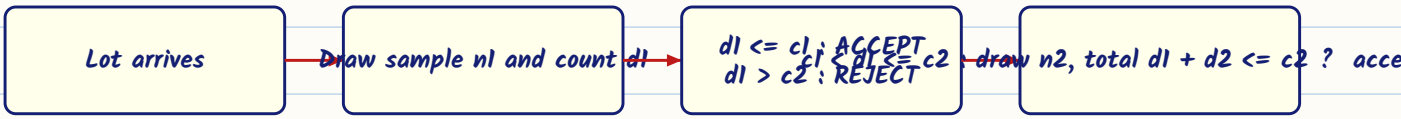
* Decision to ACCEPT or REJECT a lot is taken from the inspection of a random SAMPLE. Plans are defined by sample size n and acceptance number c (max defectives allowed in sample). Lot is accepted if defectives found <= c, otherwise rejected (or 100% inspected).

Single sampling plan

* Take ONE sample of size n from lot size N. Count defectives d. If d <= c -> accept lot; if d > c -> reject. Simple, easy to administer, but sample size (and inspection cost) is large and gives only one chance.

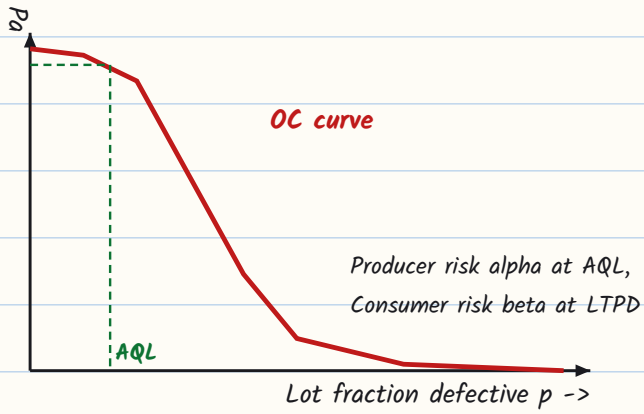
Double sampling plan

* Plan (n1, c1, n2, c2): take FIRST sample n1, with d1 defectives. If d1 <= c1 -> accept; if d1 > c2 -> reject; if c1 < d1 <= c2 -> take SECOND sample n2 with d2: if d1 + d2 <= c2 -> accept, else reject. Gives a second chance to the lot, average inspection is less than single sampling for very good / very bad lots, psychologically more acceptable, but is more complex.



Single sampling	Double sampling
One sample	Up to two samples
Larger sample, higher inspection cost	Lower average sample number (ASN)
Simple administration	More complex records
One decision chance	Second chance for borderline lots

* MULTIPLE (sequential) sampling: units tested one by one until decision. OC CURVE (Operating Characteristic): plot of probability of acceptance Pa against lot fraction defective p; shows plan's discriminating power. AQL = Acceptable Quality Level (good lot, producer risk alpha ~ 5%); LTPD / RQL = Lot Tolerance Percent Defective (bad lot, consumer risk beta ~ 10%); AOQ = average outgoing quality, AOQL = its maximum.



Pa (single plan, Poisson) = sum from d = 0 to c of e^{-np} $(np)^d$ / d!

Eg: n = 50, c = 1, p = 2%: np = 1.0; Pa = e^{-1} (1 + 1) = 0.7358 (73.6% chance of acceptance).

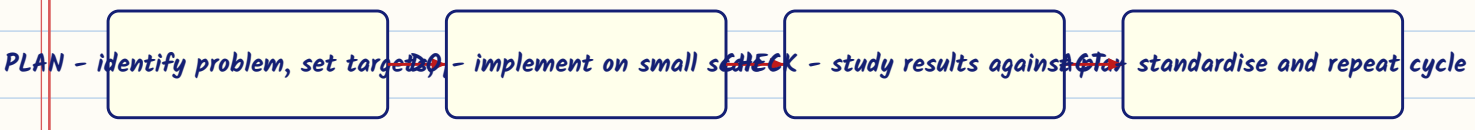
6. Total Quality Management (TQM)

* TQM = a management philosophy and system in which ALL members of the organisation continuously improve processes, products, services and culture to meet or exceed CUSTOMER expectations. 'Total' - everyone and every function; 'Quality' - meeting customer requirements; 'Management' - leadership and systems. Pioneers: W. Edwards Deming (14 points, PDCA cycle), Joseph Juran (trilogy), Philip Crosby (zero defects), Ishikawa (quality circles, fishbone), Feigenbaum (TQC), Taguchi (loss function).

Elements (principles) of TQM

* 1. CUSTOMER focus (internal and external). 2. LEADERSHIP and top management commitment. 3. EMPLOYEE involvement, empowerment and teamwork; training. 4. PROCESS approach. 5. CONTINUOUS IMPROVEMENT (Kaizen). 6. Fact based decisions - data, SPC. 7. SUPPLIER partnership. 8. Quality cost, benchmarking, quality function deployment (QFD), FMEA, JIT, 5S.

Continuous improvement - PDCA / Kaizen



* KAIZEN = small, continuous, incremental improvement by everybody (Japanese: 'change for the better'). Tools: the 7 QC tools - check sheet, histogram, Pareto diagram (80:20 rule), cause-and-effect (Ishikawa / fishbone) diagram, scatter diagram, control chart, stratification / flow chart. Also 5S (Seiri sort, Seiton set in order, Seiso shine, Seiketsu standardise, Shitsuke sustain).

Zero defect concept

* Philip B. Crosby (1960s, Martin Marietta): 'Quality is free'; performance standard = ZERO DEFECTS - do it right the FIRST time (cost of quality = cost of non-conformance). Four absolutes: quality = conformance to requirements; prevention, not appraisal; performance standard zero defects; measurement = price of nonconformance. Motivation by commitment, awareness, error cause removal and ZD Day. Related: Poka-Yoke (mistake proofing, Shigeo Shingo).

Quality circles

* A small group (6 - 12) of employees from the same work area who meet VOLUNTARILY and regularly (about 1 hour per week) to identify, analyse and solve work related quality problems and present solutions to management. Originated in Japan (Ishikawa, 1962, JUSE); in India, BHEL (1981). Structure: members, leader, facilitator, steering committee. Tools: brainstorming, Pareto, fishbone, data collection.

Benefits: employee involvement and motivation, problem solving, cost reduction, improved communication and teamwork.

Implementation of TQM and applications

* Steps: top management commitment and quality policy -> create quality council -> assess current status -> training of all employees -> select projects, form teams -> measure and benchmark -> implement improvements -> reward recognition -> review and sustain. Obstacles: lack of commitment, resistance to change, poor training, short-term thinking, no customer focus.
* Applications: manufacturing (automobiles - Toyota), services (banks, hospitals), education, IT, government, hotels. Benefits: higher customer satisfaction, lower defects and cost, market share, employee morale.

7. ISO Quality Systems

* ISO (International Organisation for Standardisation, Geneva, 1947) - ISO 9000 family: standards for QUALITY MANAGEMENT SYSTEMS (QMS). ISO 9000 - fundamentals and vocabulary; ISO 9001:2015 - requirements (the one certified); ISO 9004 - performance improvement guidelines; ISO 19011 - auditing.
* ISO 9001:2015 clauses (High Level Structure): 4 Context of the organisation, 5 Leadership, 6 Planning (risk based thinking), 7 Support, 8 Operation, 9 Performance evaluation (audit, review), 10 Improvement. Seven QM principles: customer focus, leadership, engagement of people, process approach, improvement, evidence-based decision making, relationship management.
* Other: ISO 14001 (Environmental Management), ISO 45001 (Occupational Health and Safety), ISO 27001 (Information Security),

ISO/TS 16949 -> IATF (automotive), ISO 13485 (medical devices), ISO 22000 (food safety). Certification by an accredited body after audit; surveillance audits each year; re-certification every 3 years. Benefits: documented procedures, international recognition, customer confidence, fewer errors, continual improvement.

8. Six Sigma

* SIX SIGMA = a data driven, disciplined methodology and philosophy for eliminating DEFECTS and VARIATION in any process, aiming at near-perfect quality: 3.4 DEFECTS PER MILLION OPPORTUNITIES (DPMO). Developed at MOTOROLA (Bill Smith, Mikel Harry, 1986); popularised by General Electric (Jack Welch).
* Meaning of sigma level: the process mean is 6 standard deviations from the nearest specification limit (with allowance for a 1.5 sigma long term shift of the mean).

Sigma level	DPMO	Yield %
1	691,462	30.9
2	308,538	69.1
3	66,807	93.3
4	6,210	99.38
5	233	99.977
6	3.4	99.99966

DPMO = (number of defects / (number of units x opportunities per unit)

Basic concepts

* DMAIC (improve existing process): DEFINE the problem and customer

needs (project charter, VOC); MEASURE current performance (data collection, baseline sigma); ANALYSE root causes (fishbone, Pareto, hypothesis tests, regression); IMPROVE (solutions, DOE, pilot); CONTROL (SPC, control plan, standardise). DMADV / DFSS (new process design): Define, Measure, Analyse, Design, Verify.

* Belts (roles): Champion (top executive sponsor), Master Black Belt (coach), BLACK BELT (full time project leader), GREEN BELT (part time, team member), Yellow / White Belt. Key terms: CTQ (Critical To Quality), VOC (Voice of Customer), process capability Cp / Cpk, SIPOC, defect, opportunity, yield, rolled throughput yield. LEAN SIX SIGMA = Lean (remove waste) + Six Sigma (reduce variation).

Cp = (USL - LSL) / (6 sigma) ; Cpk = min((USL - mean)/(3 sigma), (m

Revision: SQC = control charts (process) + acceptance sampling (product). X-bar limits = X-dbar +- A2 R-bar; R chart: D3, D4. TQM = customer focus + continuous improvement. Six Sigma = 3.4 DPMO, DMAIC. ISO 9001 = QMS.

UNIT - IV : FINANCIAL MANAGEMENT

I. Scope and nature of financial management

* FINANCIAL MANAGEMENT = planning, organising, directing and controlling the financial activities of an enterprise - procurement and effective utilisation of funds. (Solomon: the efficient use of an important economic resource, namely capital funds.) Objective: PROFIT maximisation (short term) and WEALTH maximisation (long term - maximising the market value of the shareholders' equity) with proper risk balance.

Scope - the three decisions



* Nature: it is a part of overall management; continuous; deals with money (cash flows), risk and return trade-off; both an ART and a SCIENCE; provides inputs for all functional areas; involves forecasting, planning, control, and reporting; applies to all types of business.
* Functions of the finance manager: estimating fund requirements, raising funds at least cost, capital structure decision, investment (fixed assets and working capital) decisions, dividend policy, cash

management, financial control and analysis (ratios), tax planning, protecting the funds (insurance, risk).

2. Sources of Finance

Basis	Sources
LONG TERM (> 5 years)	Equity (ordinary) shares, preference shares, debentures / bonds, long-term loans from banks / financial institutions (IDBI, SIDBI, IFCI), retained earnings (ploughing back), venture capital, lease financing, public deposits (3 yr)
MEDIUM TERM (1 - 5 yr)	Term loans, lease and hire purchase, public deposits, medium term debentures, commercial paper, loan from commercial banks
SHORT TERM (< 1 yr)	Trade credit, bank overdraft, cash credit, bills discounting, factoring, commercial paper, advances from customers, accrued expenses, short term loans

Internal sources	External sources
Retained earnings, depreciation funds, reserves, sale of assets	Shares, debentures, loans, deposits, trade credit, lease, venture capital, IPO

Features of important sources

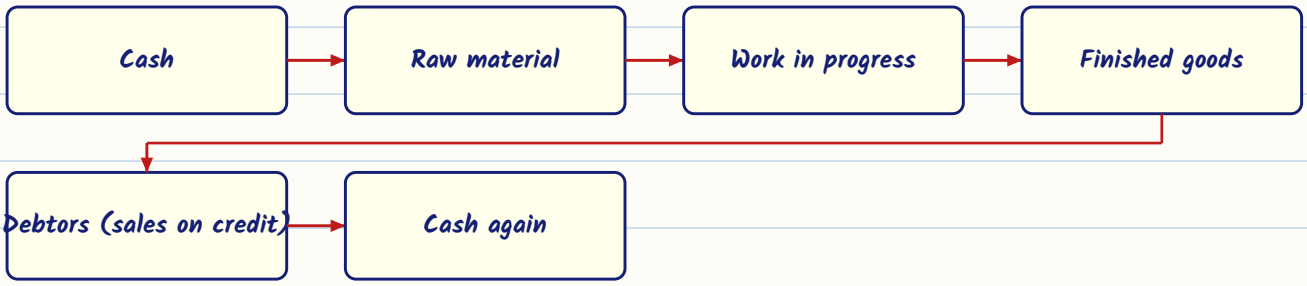
* EQUITY SHARES: owners' capital, ownership and voting rights, no fixed dividend (residual claim), no repayment (permanent), high risk and cost. PREFERENCE SHARES: fixed dividend, preference over equity in dividend and repayment, usually no voting rights; cumulative / non-cumulative, redeemable / irredeemable, convertible.
* DEBENTURES: loan capital, fixed interest, repayable on maturity,

interest is tax-deductible, no ownership dilution; secured / unsecured, convertible / non-convertible. RETAINED EARNINGS: cheapest internal source, no flotation cost. TERM LOANS: from banks and financial institutions, fixed repayment schedule. LEASE: use of asset without buying (operating and finance lease). VENTURE CAPITAL: funds for risky start-ups in exchange for equity.

Equity	Debt
Ownership, voting	No ownership
Dividend not mandatory	Interest is compulsory
No repayment	Repayable
Dividend NOT tax deductible	Interest IS tax deductible
Higher cost of capital	Lower cost, but financial risk

3. Management of Working Capital

* WORKING CAPITAL = capital needed for day to day operations. GROSS working capital = total current assets. NET working capital = current assets - current liabilities. Current assets: cash, bank, marketable securities, debtors (receivables), bills receivable, stock (raw materials, WIP, finished goods), prepaid expenses. Current liabilities: creditors, bills payable, outstanding expenses, short term loans, provisions.
* Types: permanent (regular / fixed) working capital - minimum level always needed, and temporary (variable / seasonal) working capital - fluctuates with seasons and demand.



The OPERATING CYCLE (working capital cycle) = time from cash paid for raw material to cash collected from customers:

Operating cycle = RM holding period + WIP period + FG holding period + D

Cash cycle = Operating cycle - Creditors payment period

Importance and management

* Too little working capital -> stock-outs, inability to pay creditors, loss of goodwill. Too much -> idle funds, low return, risk of theft / obsolescence. The manager balances PROFITABILITY and LIQUIDITY.
* Management techniques: CASH management (cash budget, Baumol model, float); RECEIVABLES management (credit policy, credit period, discounts, ageing schedule); INVENTORY management (EOQ, ABC analysis, JIT, safety stock, reorder level); PAYABLES management (credit terms); short term financing (bank credit, commercial paper). Current ratio = CA / CL (ideal 2:1); Quick ratio = (CA - stock)/CL (ideal 1:1).

Estimation of working capital requirements

* Method: estimate the investment in each current asset for the projected activity level, subtract current liabilities, and add a

CONTINGENCY margin (10 - 20%).

Working capital = Current assets (stock, debtors, cash, WIP) - Current li

Worked example: Annual sales (all on credit) = 12,00,000; cost of production = 10,00,000 of which raw material 4,00,000, labour 3,00,000, overheads 3,00,000. Raw material stock 1 month, WIP 0.5 month, finished goods 1 month, debtors 1.5 months, creditors 1 month, wages / overheads paid 1 month in arrears, cash balance 40,000, contingency 10%.

Item	Calculation	Amount (Rs)
Raw material stock	4,00,000 x 1/12	33,333
Work in progress	10,00,000 x 0.5/12	41,667
Finished goods	10,00,000 x 1/12	83,333
Debtors (at cost)	10,00,000 x 1.5/12	1,25,000
Cash balance	given	40,000
Total current assets		3,23,333
Creditors	4,00,000 x 1/12	33,333
Outstanding wages and overheads	6,00,000 x 1/12	50,000
Total current liabilities		83,333
Net working capital	3,23,333 - 83,333	2,40,000
Add 10% contingency		24,000
WORKING CAPITAL REQUIRED		2,64,000

4. Budget and Budgetary Control

* BUDGET = a detailed plan of operations for a future period expressed

in quantitative / monetary terms (CIMA: a financial and / or quantitative statement prepared prior to a defined period of time of the policy to be pursued to attain a given objective). BUDGETARY CONTROL = establishing budgets, continuously comparing ACTUAL with BUDGETED results, finding VARIANCES and taking corrective action.

Objectives and steps

* Planning and coordination, control of costs, communication, motivation, performance evaluation, fixing responsibility, optimum use of resources. Steps: set organisation objectives -> budget committee and budget manager -> identify the LIMITING (principal budget) factor (usually sales) -> prepare budgets -> coordinate and approve -> implement -> compare actual with budget -> analyse variances -> corrective action.

Types of budgets

Type	Examples / meaning
Functional budgets	Sales, production, raw material purchase, labour, overhead, cost of goods sold, capital expenditure, research budget
Master budget	Summary of all functional budgets: budgeted P&L account and balance sheet
Fixed budget	Prepared for ONE level of activity, not changed
Flexible budget	Adjusted for different activity levels (variable cost changes with output)
Cash budget	Estimated cash receipts and payments
Zero base budget (ZBB)	Every item justified afresh from zero, ignoring past
Performance / programme budget	Based on programmes and expected results

* Advantages: discipline in planning, coordination, cost consciousness, early warning, goal congruence. Limitations: based on estimates, rigid if not flexible, time consuming and costly, resistance from managers, cannot replace management.

5. Capital Budgeting

* CAPITAL BUDGETING = planning, evaluating and selecting long term investment proposals (acquiring fixed assets - machinery, new plant, new product, expansion, replacement) whose benefits are expected over a number of years. Also called capital expenditure decision / investment appraisal.

Nature of investment decisions

* Large funds committed; LONG term effect; IRREVERSIBLE (high cost of reversal); strategic - affects growth, profitability and risk of the firm; involves uncertainty about future cash flows; types: expansion, replacement and modernisation, diversification, independent / mutually exclusive / contingent projects; decisions need accurate cash flow forecasting (after tax, incremental, including depreciation tax shield and working capital).

Investment evaluation criteria

Non-discounting (traditional)	Discounting (time value of money)
Payback period (PBP)	Net Present Value (NPV)
Accounting rate of return (ARR)	Internal Rate of Return (IRR)
	Profitability Index (PI) / Benefit - cost ratio

(1) Payback period (PBP)

* Time required to recover the initial investment from net cash inflows. Accept if PBP < standard period. Merits: simple, favours liquidity, risk indicator. Demerits: ignores time value of money and cash flows after payback; ignores profitability.

PBP (even cash flows) = Initial investment / Annual cash inflow

PBP (uneven) = Years before full recovery + (Unrecovered amount / Cash

(2) Accounting (Average) Rate of Return (ARR)

ARR = (Average annual profit after tax / Average (or initial) investment)

* Average investment = (Initial investment + Salvage value) / 2. Accept

if ARR > required rate. Uses accounting profit, not cash flow; ignores time value.

(3) Net Present Value (NPV)

$$NPV = \text{sum from } t = 1 \text{ to } n \text{ of } C_t / (1 + k)^t - I_0$$

* C_t = cash inflow in year t, k = discount rate (cost of capital), I_0 = initial investment. DECISION: accept if NPV > 0; reject if NPV < 0; among mutually exclusive projects choose the highest NPV. Merits: time value, whole life, maximises wealth. Demerit: needs cost of capital; absolute measure.

(4) Internal Rate of Return (IRR)

* The discount rate at which NPV = 0 (present value of inflows = outflow). Found by trial and error / interpolation:

$$IRR = r_1 + [\ NPV_1 / (NPV_1 - NPV_2)\ ] \times (r_2 - r_1)$$

* Accept if IRR > cost of capital (k). Merits: percentage, time value. Demerits: multiple IRRs for non-conventional flows; may conflict with NPV for mutually exclusive projects (assumes reinvestment at IRR).

(5) Profitability Index (PI)

$$PI = \text{Present value of cash inflows} / \text{Initial investment} = 1 + NPV / I_0$$

* Accept if PI > 1. Useful for ranking projects under capital rationing.

Numerical problem

A project needs an investment of Rs 1,00,000 and gives net cash inflows of Rs 40,000, 40,000, 40,000 and 30,000 in years 1 - 4. Cost of capital = 10%. Scrap value nil. Evaluate by all methods.

Year	Cash flow	PV factor 10%	PV (Rs)	Cumulative CF
0	-1,00,000	1.000	-1,00,000	-1,00,000
1	40,000	0.909	36,360	-60,000
2	40,000	0.826	33,040	-20,000
3	40,000	0.751	30,040	+20,000
4	30,000	0.683	20,490	+50,000

* PBP: recovered during year 3: 2 + 20,000/40,000 = 2.5 years.
* NPV = (36,360 + 33,040 + 30,040 + 20,490) - 1,00,000 = 1,19,930 - 1,00,000 = + Rs 19,930 -> ACCEPT.
* PI = 1,19,930 / 1,00,000 = 1.20 (> 1) -> accept.
* ARR: total inflow = 1,50,000, net profit over life (before depreciation treatment) = 50,000 -> average annual profit = 12,500 -> ARR on initial investment = 12,500/1,00,000 = 12.5%; on average investment (50,000) = 25%.
* IRR: try 18%: PV = 40,000 (0.847 + 0.718 + 0.609) + 30,000 (0.516) = 86,960 + 15,480 = 1,02,440 -> NPV = +2,440. Try 20%: 40,000 (0.833 + 0.694 + 0.579) + 30,000 (0.482) = 84,240 + 14,460 = 98,700 -> NPV = -1,300.

IRR = 18 + [2,440 / (2,440 + 1,300)] x 2 = 18 + 1.30 = 19.3% (> 10%, ac

Second example - choosing between two projects

	Project A	Project B
Investment	Rs 50,000	Rs 50,000
Cash inflows yr 1 to 3	25,000 each	10,000, 25,000, 40,000
PBP	2.0 years	2.4 years
NPV at 10%	25,000 x 2.487 - 50,000 = +12,175	9,090 + 20,650 + 30,040 - 50,000 = +9,780
PI	1.24	1.20
Decision	A is preferred (higher NPV, faster payback)	

Criterion	Accept if	Time value?
PBP	PBP < standard	No
ARR	ARR > required rate	No
NPV	NPV > 0	Yes
IRR	IRR > cost of capital	Yes
PI	PI > 1	Yes

Time value of money (basics)

Future value FV = P (1 + r)^n ; Present value PV = F / (1 + r)^n ; PV of ann

* A rupee today is worth more than a rupee tomorrow (inflation, opportunity cost, risk). Compounding moves money forward in time; discounting brings future cash flows to the present.

Revision: 3 finance decisions - investment, financing, dividend. WC = CA - CL. Operating cycle. Budget types: fixed, flexible, ZBB. Capital budgeting: PBP, ARR (non-discounting); NPV > 0, IRR > k, PI > 1 (discounting).

UNIT - V : HUMAN RESOURCE MANAGEMENT AND VALUE

I. Concept of Human Resource Management (HRM)

* HRM = the management function concerned with PLANNING, ACQUIRING (recruiting, selecting), DEVELOPING, MOTIVATING, COMPENSATING, MAINTAINING and integrating the human resources of an organisation so that individual, organisational and social objectives are achieved. People are treated as a valuable ASSET (human capital), not just a cost.
* Features: pervasive, people oriented, continuous, developmental, strategic, a staff (advisory) function, based on human values, integrates employee and organisation goals.

Personnel management vs HRM

Personnel management	Human resource management
Traditional, administrative, routine	Modern, strategic, proactive
Employees are a cost / a tool	Employees are an asset / resource
Focus on rules, procedures, records	Focus on development, commitment, culture
Short term, reactive	Long term, planned
Personnel dept. alone responsible	Responsibility of all line managers also
Maintaining and welfare of staff	Growth, empowerment of staff

Industrial relations (IR)

* Relationship between EMPLOYERS (management), EMPLOYEES (workers) and their TRADE UNIONS, and the government, in the industrial setting. Aims: harmonious relations, industrial peace, higher productivity. Components: collective bargaining, trade unions, joint

consultation, grievance handling, discipline, workers' participation in management, dispute settlement.

* Industrial disputes: strikes, lock-outs, gherao, go-slow. Machinery: Works committee, conciliation officer, board of conciliation, court of inquiry, labour court, tribunal, arbitration. Laws: Industrial Disputes Act 1947, Trade Unions Act 1926, Factories Act 1948, Minimum Wages Act 1948, Payment of Wages, Bonus, Gratuity, ESI and Provident Fund Acts (now being merged into labour codes).

2. Functions of Personnel Management

Type	Function	Activities
MANAGERIAL	Planning	Manpower planning, policies, budgets, forecasting
	Organising	Structure, job design, authority relations
	Directing	Motivation, leadership, communication
	Controlling	Audits, performance review, cost control
OPERATIVE	Procurement	Job analysis, human resource planning, recruitment, selection, placement, induction
	Development	Training, career planning, management development
	Compensation	Job evaluation, wages and salary, incentives, benefits
	Integration	Motivation, grievance handling, industrial relations, union dealing
	Maintenance	Health, safety, welfare, social security
	Separation	Retirement, resignation, retrenchment, VRS

* JOB ANALYSIS gives JOB DESCRIPTION (duties, responsibilities, working conditions) and JOB SPECIFICATION (qualifications, skills, experience needed by the worker). Recruitment = searching for candidates (internal / external); selection = choosing the best (tests, interviews).

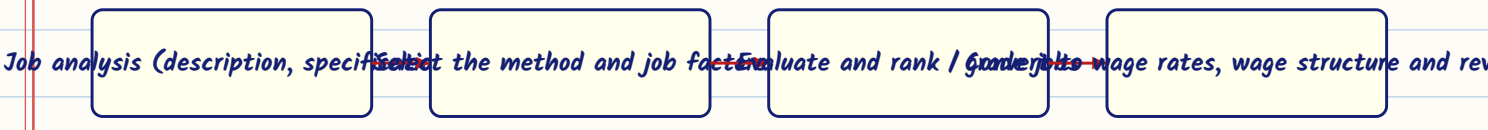
3. Job Evaluation

* JOB EVALUATION = systematic process of determining the RELATIVE WORTH (value) of jobs in an organisation, in order to establish a fair and equitable WAGE and SALARY structure. It rates the JOB (its duties, skill, responsibility, effort, working conditions), NOT the person who does it.

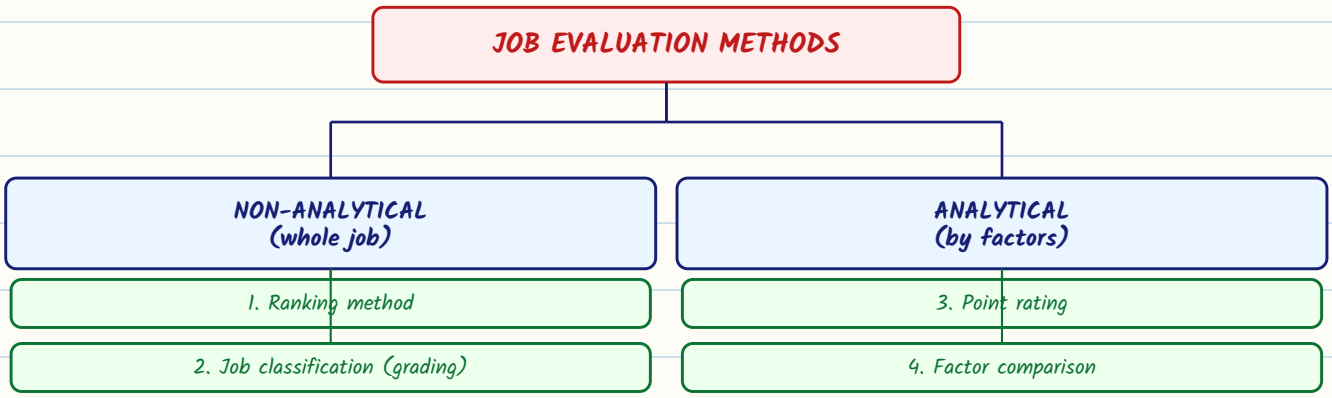
Importance / objectives

* Basis of fair and logical wage differentials; removes pay inequity and grievances; helps in recruitment (fixing starting pay), promotion, transfers; establishes a hierarchy of jobs; assists in wage negotiations and in comparing wages with other firms; clarifies job duties and organisation structure; reduces labour unrest.

Procedure



Types (methods) of job evaluation



Method	Procedure	Merits / Demerits
1. RANKING	Jobs are arranged in order of importance (highest to lowest) by comparing each job with others, using a committee; rank -> pay	Simple, quick, cheap; subjective, only for small firms; no measure of the gap between jobs
2. JOB CLASSIFICATION (grading)	Predetermined grades / classes with descriptions (Grade I - unskilled ... Grade V - skilled supervisor); each job is slotted into a grade	Simple, used in government; difficult to write class descriptions, rigid
3. POINT RATING (point system)	Select job FACTORS (skill, effort, responsibility, working conditions), divide into degrees, give POINTS, add points for each job; convert total points to money	Most widely used, systematic, reliable; costly and time consuming, factor selection is tricky
4. FACTOR COMPARISON	Key (benchmark) jobs are ranked factor by factor and their wages are split among the factors; other jobs are compared and slotted money-wise	Direct money values, objective; complex, key job wages may be wrong

Point rating example: Factors and maximum points - Skill 40, Effort 20, Responsibility 25, Job conditions 15 = 100. A job scoring 30 + 12 + 20 + 8 = 70 points; if one point = Rs 200 -> basic pay Rs 14,000.

4. Merit Rating (Performance Appraisal)

* MERIT RATING = systematic evaluation of the PERSONAL QUALITIES

and PERFORMANCE of an EMPLOYEE (not the job) by superiors, against defined standards, usually once or twice a year. Job evaluation rates the job; merit rating rates the man doing it.

* Objectives: basis for increments, bonus, promotion, transfer; identifies training needs; improves performance by feedback; helps in selection validation; motivates employees; improves supervisor - subordinate relations.

Job evaluation	Merit rating
Rates the JOB	Rates the EMPLOYEE (performance)
Fixes basic wage / grade of the job	Decides increments, bonus, promotion
Done when job is created or changed	Done periodically (yearly)
Based on job factors (skill, effort ...)	Based on traits (quality, initiative, reliability, attendance)

Quantitative / rating methods of merit rating

* 1. RANKING method: employees are arranged from best to worst. 2. PAIRED COMPARISON: each employee compared with every other, one pair at a time; number of comparisons = n(n - 1)/2. 3. GRAPHIC RATING SCALE: traits (quality of work, quantity, dependability, initiative, cooperation, attendance) rated on a scale (e.g. 1 - 5 or poor to excellent); scores added - the most common. 4. FORCED CHOICE (distribution): rater picks from statements describing the employee; forced distribution (bell curve: 10% outstanding, 20% good, 40% average ...). 5. CHECKLIST: yes / no questions with weights. 6. CRITICAL INCIDENT: record of exceptionally good or bad incidents.
* 7. FIELD REVIEW, essay (descriptive) appraisal, MBO (management by objectives - achievement vs goals), 360 degree feedback (superiors, peers, subordinates, customers), BARS (behaviourally anchored rating

scale), assessment centres.

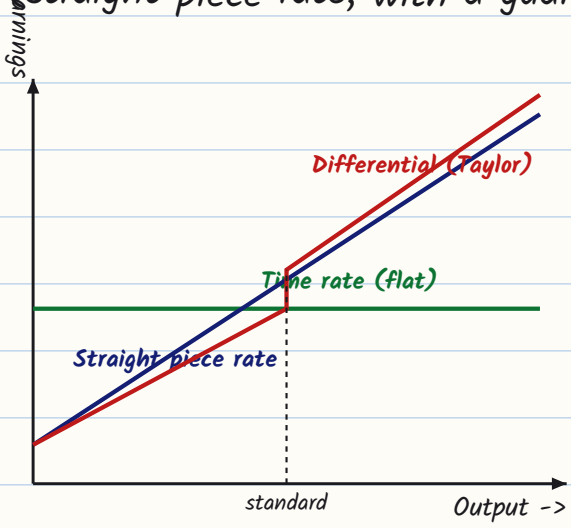
* Errors: halo effect, central tendency, leniency / strictness, personal bias, recency effect, spill-over of past performance.

5. Wage Incentive Plans

* WAGE = payment for labour. Wage incentive = extra payment linked to output / performance above a standard to MOTIVATE workers to higher productivity. Characteristics of a good plan: simple, fair, easy to understand, accurate standards (time study), guaranteed minimum wage, attainable, benefits both employer and employee, quick payment, no cost increase beyond productivity gain.

Basic wage systems

* TIME RATE: pay = hours worked x rate per hour (wages paid irrespective of output; simple but no incentive; used where output is not measurable). PIECE RATE: pay = number of pieces x rate per piece (straight piece rate, with a guaranteed minimum).



Types of incentive plans

Plan	Description	Formula / example
TAYLOR'S differential piece rate	Two piece rates: HIGH rate (e.g. 120%) for output at or above standard, LOW rate (e.g. 80%) below standard; penalises slow workers	Std 100 pcs at Rs 10: 110 pcs earn 110 x 12 = 1320; 90 pcs earn 90 x 8 = 720
MERRICK multiple piece rate	Three rates: up to 83% of standard - ordinary rate; 83 - 100% - 110% of rate; above 100% - 120%	Milder than Taylor's
HALSEY premium plan (50 : 50)	Time saved is shared: worker gets a fixed % (usually 50%) of the time saved at hourly rate	W = T x R + p (S - T) R
ROWAN plan	Bonus = time saved / time allowed of the time taken; bonus rises at decreasing rate (cannot exceed the wage)	W = T x R + [(S - T)/S] T R
GANTT task and bonus	Day rate guaranteed; if task completed in std time, bonus 20% of wage; at or above standard worker earns 120% of piece rate	Efficiency based
EMERSON efficiency	Bonus starts at 67% efficiency, rising to 20% at 100% efficiency and 1% for each further 1%	Efficiency %
BEDAUX point plan	Work measured in B (point) units - 1 B = 1 minute of work + rest; premium bonus on excess B per hour	Standard 60 B per hour
Group incentives / profit sharing / bonus	Group earns extra for team output; Scanlon plan (cost saving share), Rucker plan, profit sharing, co-partnership, ESOP	Encourages team work

T = time taken, S = time allowed (standard), R = hourly wage rate, p = share of time saved.

Eg: S = 10 h, T = 8 h, R = Rs 50. Halsey (50%): W = 8 x 50 + 0.5 (2) 50 = 400 + 50 = Rs 450. Rowan: W = 400 + (2/10) x 8 x 50 = 400 + 80 = Rs 480. Time wage = Rs 400 (guaranteed, since T x R).

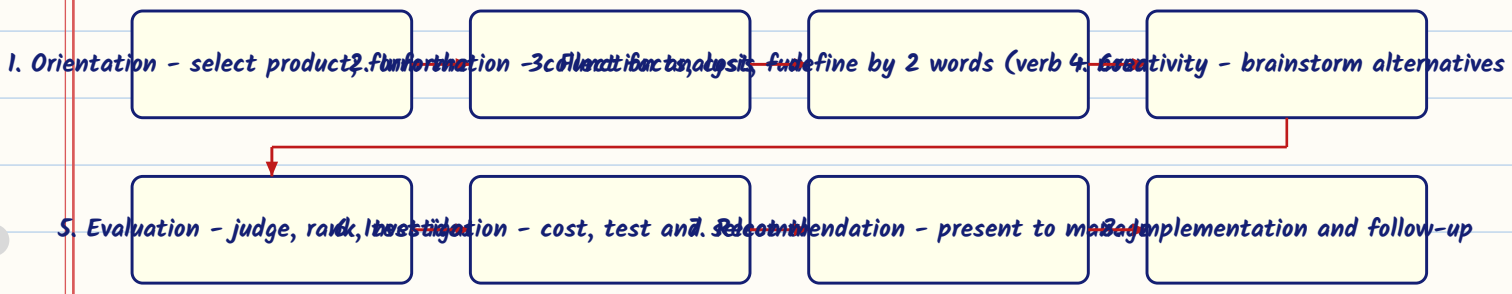
6. Value Analysis and Value Engineering

* VALUE = FUNCTION / COST (worth of the function to the customer divided by cost to provide it). Value increases when function improves at same cost, or same function at lower cost. Developed by LAWRENCE D. MILES at General Electric (USA) in 1947 - 49 as 'Value Analysis' when materials were scarce.
* VALUE ANALYSIS (VA) = organised, creative approach for identifying UNNECESSARY COST in an EXISTING product / process and removing it without reducing quality, performance, reliability or appearance. VALUE ENGINEERING (VE) = same techniques applied at the DESIGN stage of a NEW product (prevention rather than cure).

Types of value

* USE value (utility / performance), ESTEEM (prestige) value (attractiveness), COST value (sum of labour, material, overhead), EXCHANGE value (market price). Functions: PRIMARY (basic - why the product exists) and SECONDARY (supporting). Eg: a pen - basic function: make marks; secondary: appearance, clip.

Implementation procedure - job plan

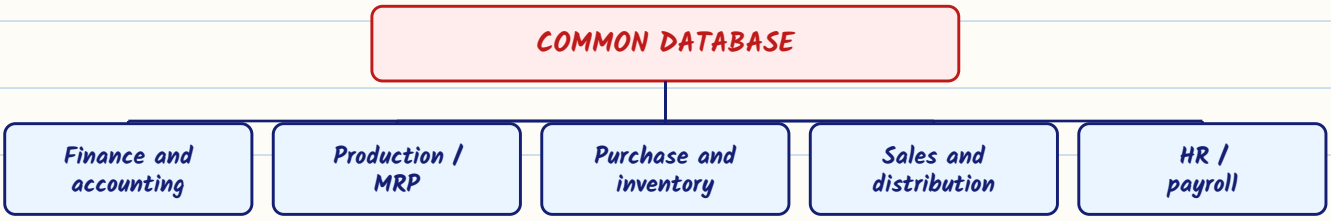


* Questions of VA (Miles): What is it? What does it do? What does it cost? What else will do the job? What does that cost? Targets of VA: high cost, large volume, many parts, complex, old design. Techniques: ABC analysis, FAST (Function Analysis System Technique) diagrams, Pareto, DARSIRI method, brainstorming, Delphi, make-or-buy, standardisation, cost benefit analysis, value index.
* Benefits: reduces cost (10 - 30% typical), improves quality and performance, removes unnecessary features, uses new materials and processes, better customer value, teamwork. Eg: replacing a machined bracket by a stamped one, or brass by plastic bush giving the same function at lower cost.

7. Enterprise Resource Planning (ERP)

* ERP = an integrated, computer based software system that manages and links ALL major business functions - finance, accounting,

production, inventory, purchase, sales, HR, CRM - through a single database and common platform, so that information flows in real time across the whole enterprise. Evolved from MRP (Material Requirement Planning) -> MRP II (Manufacturing Resource Planning) -> ERP (1990s). Vendors: SAP, Oracle, Microsoft Dynamics, Infor, Tally.

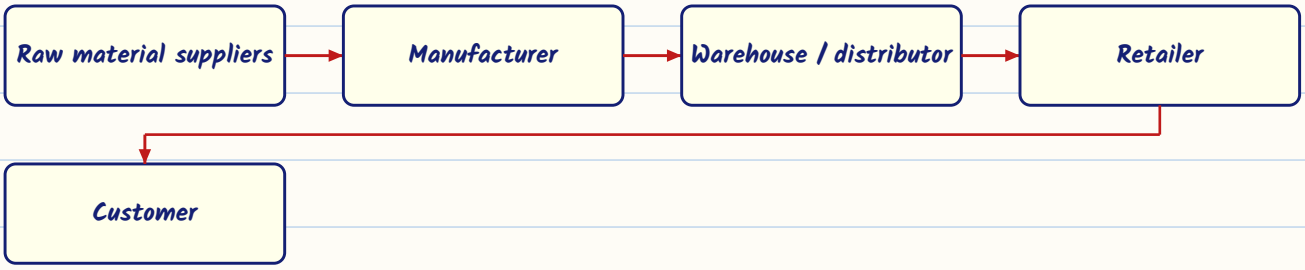


* Features: integration, single database, real time data, modular (plug in), automation of processes, standard best-practice workflow, customisable, access control, reporting and analytics.
* Benefits: better information and decisions, lower inventory and cost, shorter cycle time, improved customer service, avoids duplicate data, productivity, transparency, regulatory compliance. Limitations: very high cost, long implementation, resistance to change, training need, customisation and data migration problems, vendor dependence. Implementation phases: planning and vendor selection, business process re-engineering (BPR), configuration, data migration, testing, training, go-live, support.

8. Supply Chain Management (SCM)

* SUPPLY CHAIN = network of organisations, people, activities, information and resources involved in moving a product from

SUPPLIERS to CUSTOMERS: suppliers -> manufacturer -> distributors / wholesalers -> retailers -> customers, with flows of material, information and money (the reverse flow = returns / reverse logistics).



* SCM = planning, coordinating and controlling the whole network of suppliers, plants, warehouses and customers to deliver the right product, in the right quantity, at the right place and time, at least total cost, with high customer satisfaction.
* Components: PLAN (demand forecasting, production and inventory plan), SOURCE (supplier selection, procurement, e-procurement), MAKE (manufacturing, quality), DELIVER (logistics, warehousing, transportation, order management), RETURN (reverse logistics) - the SCOR model. Drivers: facilities, inventory, transportation, information, sourcing, pricing.
* Key concepts: BULLWHIP EFFECT (small changes in consumer demand amplify as one moves upstream), just-in-time (JIT), vendor managed inventory, outsourcing, third party logistics (3PL), strategic supplier partnership, postponement, cross docking, RFID and IoT tracking, e-commerce, blockchain, green supply chain, agile / lean supply chain.
* Benefits: reduced inventory and logistics cost, better customer service, shorter lead time, fewer stock-outs, improved supplier relations, higher profit and competitive advantage. Challenges: demand uncertainty, lack of trust and information sharing, global risks, IT integration cost.

Revision: HRM = strategic people management; job evaluation rates JOB (ranking, classification, point rating, factor comparison); merit rating rates MAN; Halsey, Rowan, Taylor plans; VA/VE: value = function / cost; ERP = integrated software; SCM = supplier to customer flow.